

A novel nomogram for the prediction of perforation during endoscopic submucosal dissection for colorectal neoplasms

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Abstract

Background: High perforation risk hinders the widespread adoption of ESD for colorectal neoplasms. This study was performed to determine the risk factors of colorectal endoscopic submucosal dissection (ESD)-induced perforation and develop a predictive model.

Methods: A total of 1046 colorectal neoplasms in 1011 patients were retrospectively enrolled from January 2011 to December 2021, in a single tertiary center as the derivation cohort. We identified independent risk factors for perforation using univariate analysis and multi-variate logistic regression. A nomogram was developed based on the logistic regression model and prospectively applied to 266 colorectal neoplasms as the validation cohort. The performance of the predictive model was evaluated with the receiver operating characteristic curve, calibration plot, and decision curve analysis.

Results: Independent pre-operative factors for colorectal ESD-induced perforation were tumor located in the left colon [odds ratio (OR) 2.39, $P = 0.040$], size ≥ 40 mm (OR 3.36, $P < 0.001$), $\geq 2/3$ circumference (OR 7.55, $P = 0.004$), located across folds (OR 6.26, $P < 0.001$), and laterally spreading tumor (non-granular type, OR 2.34, $P = 0.029$; granular type, OR 2.46, $P = 0.021$). The nomogram model incorporating the pre-operative factors performed well in both the derivation and validation cohorts (areas under the curve of 0.750 and 0.806, respectively). Decision curve analysis demonstrated that the clinical benefit of the nomogram was favorable.

Conclusions: The novel nomogram, developed and prospectively validated, incorporating tumor size, location, and morphology can successfully predict perforation during ESD for colorectal neoplasms.

Keywords: Colorectal cancer, endoscopic submucosal dissection, nomogram, perforation

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INTRODUCTION

Colorectal carcinoma (CRC) ranks as the second principal cause of cancer-related mortality globally.^[1] The majority

of CRC develops from colorectal adenomas or sessile serrated lesions over a prolonged period.^[2,3] Endoscopic identification and resection of such early-stage neoplasms are associated with a decreased risk of CRC-related mortality^[4] and are now recommended by many national

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guidelines.^[5] As an effective endoscopic resection technique, endoscopic submucosal dissection (ESD) provides a high en bloc resection rate, precise pathological evaluations, and a lower local recurrence rate, especially for large (≥ 20 mm) colorectal polyps.^[6-9] However, the utilization of ESD in colorectal lesions is restricted due to the high degree of technical difficulty and risk of complications. Perforation is one of the most severe complications during colorectal ESD as it can result in peritonitis or even fatal septic shock due to the contamination of colonic bacteria and feces. The reported perforation rate for colorectal ESD is significantly high, ranging from 1.4% to 20.4%.^[10-13] Several risk factors have been identified, including tumor size, submucosal fibrosis, tumor location, and the experience of the endoscopists.^[14-16] However, the available tools for predicting colorectal ESD-related perforation are currently limited. Predictive nomograms have emerged as valuable tools in clinical settings by integrating multiple variables and providing an individual numerical probability of specific clinical events.^[17] The aim of the current study, therefore, was to identify risk factors of perforation during colorectal ESD using a sizeable dataset, and subsequently develop and validate a new nomogram to predict perforation in patients with colorectal ESD.

PATIENTS AND METHODS

Patients and lesions

A total of 1011 patients with 1046 colorectal neoplasms who underwent ESD at Peking University Third Hospital, from January 2011 to December 2021, were retrospectively evaluated and enrolled in the study. Patient demographic features, lesion characteristics, procedure time, adverse events, and post-operative pathology results were collected to investigate the risk factors of ESD-related perforation, and the prediction model was constructed based on the preoperative factors. We prospectively enrolled 240 consecutive patients with 266 lesions that underwent colorectal ESD, from January 2022 to December 2022, as the validation cohort. Written informed consent was obtained from all patients. The study was approved by the ethics committee of Peking University Third Hospital (M2022769) in accordance with the ethical guidelines of the 1975 Declaration of Helsinki.

ESD procedure

The indications for colorectal ESD were in accordance with the Japan Gastroenterological Endoscopy Society guidelines, including lesions for which endoscopic en bloc resection was required but was hard to achieve with endoscopic mucosal resection (EMR), such as non-granular laterally spreading tumor (LST-NG), lesions with Vi-type

pit pattern, carcinoma with shallow T1 invasion, large depressed tumors, large protruded lesions suspected to be malignant, or those with submucosal fibrosis.^[18] After bowel preparation, patients underwent the procedure under general anesthesia, with endotracheal intubation and carbon dioxide insufflation. Lesions were carefully examined and recorded before resection. Submucosal injection was performed using a solution of adrenaline (1:20,000) and a methylene blue mixture to provide a submucosal cushion. After the injection, the mucosa around the raised lesion was incised as circumferential incisions. Then submucosal dissections were performed using a dual or IT knife (Olympus, Tokyo, Japan) until the lesion was completely resected from the muscle layer. Hybrid ESD was defined as the combined use of endoscopic resection and the ESD technique, with the final resection accomplished through snaring.^[19]

Data collection and definition

The clinicopathologic information collected for each patient and lesion included age at the time of procedure, sex, tumor location, tumor size, macroscopic type, non-lifting sign, submucosal fibrosis, procedure time, histological type, invasion depth, and endoscopist's experience. Tumor location was defined into three categories: the rectum, the left-sided colon, and the right-sided colon, according to the Japanese Classification of Colorectal Carcinoma.^[20] Tumor size was measured endoscopically by comparison with forceps diameter.^[21] The macroscopic type is based on the Paris classification, which includes protruding types (0-Is, 0-Isp, 0-Ip) and flat or depressed types (0-IIa, 0-IIb, 0-IIa+IIc), with LST defined by Kudo's classification as a flat lesion greater than 10 mm that grows laterally along the colonic wall, distinguishing it from other flat lesions.^[22,23] Lesions were located across folds identified if the lesion margin at the oral side and/or the anal side passed through a fold completely.^[24] Non-lifting sign refers to the inability of the mucosa to elevate after submucosal injection.^[25] Submucosal fibrosis was defined as the appearance of white fibers in the transparent submucosal layer and was classified according to previous reports as F0 (no fibrosis), F1 (mild fibrosis), and F2 (severe fibrosis).^[26] Procedure time was calculated from the beginning of the submucosal injection to the completion of the lesion resection. Pathologic diagnosis conformed to the Japanese classification of cancer of the colon and rectum and the Vienna classification.^[20,27] Endoscopist's experience was considered on the basis of having successfully performed 50 colorectal ESD procedures.^[28]

Perforation includes immediate perforation and delayed perforation. Intra-operative perforation is defined as

Table 1: Clinicopathologic characteristics of the derivation and validation sets

Variable	Category	No. (%)		P
		Derivation set (n=1046)	Validation set (n=266)	
Age, years	Mean	63.8	61.4	0.003*
	SD	10.8	11.9	
Sex	Female	468 (46.3)	118 (49.2)	0.422
	Male	543 (53.7)	122 (50.8)	
BMI, kg/m ²	Mean	24.1	24.4	0.279
	SD	3.3	3.5	
Tumour location	Rectum	286 (27.3)	61 (22.9)	0.157
	Left colon	234 (22.4)	54 (20.9)	
	Right colon	526 (50.3)	151 (56.8)	
Tumour size, mm	Median	21	20	0.148
	Q1-Q3	15-30	15-30	
Circumference of the lesion	< 2/3	1035 (98.9)	264 (99.2)	0.925
	≥ 2/3	11 (1.1)	2 (0.8)	
Location with respect to colonic fold	Between folds	1019 (97.4)	259 (97.4)	0.963
	Across folds	27 (2.6)	7 (2.6)	
Preoperative biopsy	No	535 (51.1)	174 (65.4)	<0.001*
	Yes	511 (48.9)	92 (34.6)	
Location with scar	No	1015 (97.0)	262 (98.5)	0.269
	Yes	31 (3.0)	4 (1.5)	
Macroscopic type	Non-LST	379 (36.2)	79 (29.7)	0.068
	LST-G	343 (32.8)	105 (39.5)	
	LST-NG	324 (31.0)	82 (30.8)	
Non-lifting sign	Negative	971 (92.8)	251 (94.4)	0.378
	Positive	75 (7.2)	15 (5.6)	
Submucosal fibrosis	F0/F1	875 (83.7)	231 (86.8)	0.202
	F2	171 (16.3)	35 (13.2)	
Tumour depth	Mucosa	947 (90.5)	241 (90.6)	0.881
	Submucosa <1000 µm	57 (5.5)	13 (4.9)	
	Submucosa ≥1000 µm	42 (4.0)	12 (4.5)	
Histological type	Adenoma	590 (56.4)	138 (51.9)	0.282
	Sessile serrated lesions	149 (14.2)	47 (17.7)	
	Carcinoma	307 (29.3)	81 (30.5)	
Endoscopist experience	<50 cases	383 (36.6)	71 (26.7)	0.002*
	≥50 cases	663 (63.4)	195 (73.3)	
ESD type	Conventional ESD	280 (26.8)	83 (31.2)	0.149
	Hybrid ESD	766 (73.2)	183 (68.8)	
Perforation	Immediate	57 (5.5)	20 (7.5)	0.308
	Delayed	11 (1.2)	2 (0.8)	
Procedure time	Median	28	28	0.591
	Q1-Q3	14-60	15-52	

SD, standard deviation; BMI, body mass index; LST, laterally spreading tumour; SM, submucosal layer; ESD, endoscopic submucosal dissection; *P< 0.05

the presence of gas or luminal contents outside the gastrointestinal tract and was diagnosed upon direct observation of mesenteric fat or the intra-abdominal space during the procedure, while delayed perforation is diagnosed based on symptoms and radiological evidence.^[29]

Statistical analysis

All analyses were performed with R 4.3.2 (The R Foundation for Statistical Computing, Vienna, Austria). Categorical variables are presented as frequencies and percentages, variables with a continuous normal distribution are described as mean ± standard deviation (SD), and non-normal variables are described as median (inter-quartile range). Categorical variables were analyzed by using the Chi-square test, and continuous variables were analyzed by Student's *t*-test or the Mann–Whitney U-test to compare the derivation cohort with the validation cohort.

Independent risk factors were identified using a multivariate logistic regression analysis via the backward stepwise technique in the derivation cohort. The same method was employed to identify independent pre-operative factors. Subsequently, significant pre-operative factors were combined to develop a nomogram model, and the best-fit model was determined with minimal Akaike information criterion (AIC) using the “rms” package in R version. The discrimination ability of the nomogram was evaluated by the receiver operating characteristic (ROC) analysis and presented as the area under the curve (AUC) using the “pROC” package in R version. Consistency between actual and nomogram-predicted generalization probabilities was assessed by calibration curves (1,000 resampling bootstraps) in both the derivation and validation cohorts. The Hosmer–Lemeshow goodness-of-fit statistic was also used to test the calibration of the model. Decision

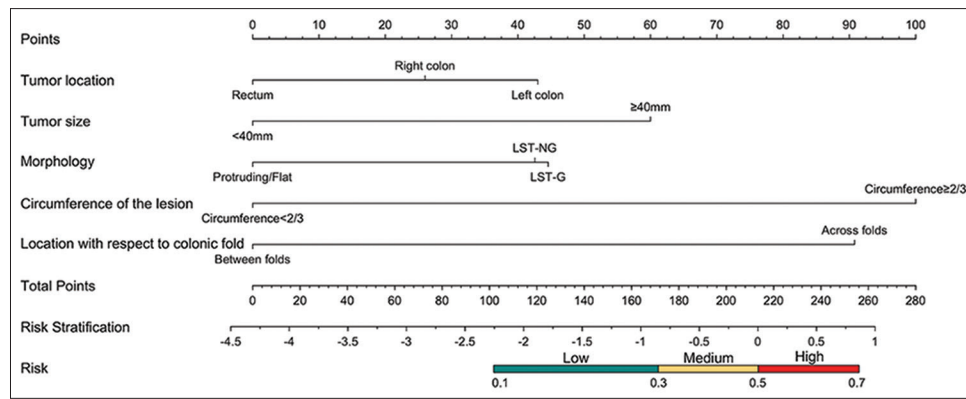


Figure 1: Nomogram for predicting colorectal ESD-induced perforation

curve analysis (DCA) was implemented to demonstrate the clinical efficacy of the predictive nomogram using the “dcurves” package in R version. A two-tailed P -value <0.05 was considered statistically significant.

RESULTS

Characteristics in the derivation and validation cohorts

A total of 1046 colorectal neoplasms performed from 2011 to 2021 were included as the derivation cohort. Two hundred and sixty six independent lesions performed during 2022 were prospectively enrolled as the validation cohort. The clinical and endoscopic characteristics of the derivation and validation cohorts are summarized in Table 1. Among the 1046 lesions in the derivation cohort, immediate and delayed perforations occurred in 57 (5.5%) and 11 (1.2%) lesions, respectively. In the validation cohort, immediate and delayed perforations occurred in 20 (7.5%) and 2 (0.8%) lesions, respectively. The rate of perforation was similar in the two cohorts (derivation, 6.7%; validation, 8.3%; $P = 0.308$). The mean age in the derivation cohort was older than that in the validation cohort ($P = 0.003$). The validation cohort had a higher proportion of pre-operative biopsy ($P < 0.001$) and endoscopist experience ($P = 0.002$). However, there were

no significant differences in other clinicopathologic factors between the cohorts.

Risk factors and pre-operative factors predicting perforation in univariate and multivariate logistic analysis

The risk factors for perforation were analyzed between the perforation group ($n = 68$) and the non-perforation group ($n = 978$) [Table 2]. Tumor size, circumference, located across folds, submucosal fibrosis, and procedure time were significantly associated with perforation (all, $P < 0.001$). Macroscopic type was also related to perforation ($P < 0.05$). Rectal lesions tended to have a lower perforation rate. There were no significant differences between the two groups in age, sex, BMI, tumor invasion depth, histological type, non-lifting sign, endoscopist's experience, or ESD type. All factors with significant differences in the univariate analysis were entered into the multivariate analysis using backward stepwise multivariate logistic regression. Additionally, the tumor location was included as well. In the logistic regression analysis, tumor location (left colon, OR = 2.96, 95% CI: 1.46-6.19, $P = 0.013$; right colon, OR = 2.19, 95% CI: 1.17-4.29, $P = 0.047$), tumor $\geq 2/3$ circumference (OR = 7.18, 95% CI: 2.18-23.44, $P = 0.006$), location across folds (OR = 3.69, 95% CI: 1.63-8.18, $P = 0.007$), submucosal fibrosis (OR = 2.52, 95% CI:

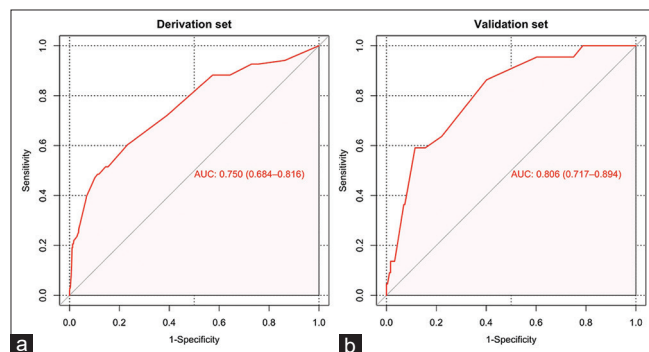


Figure 2: Receiver operating curve of the derivation set and validation set for the nomogram. a) Derivation set, b) validation set

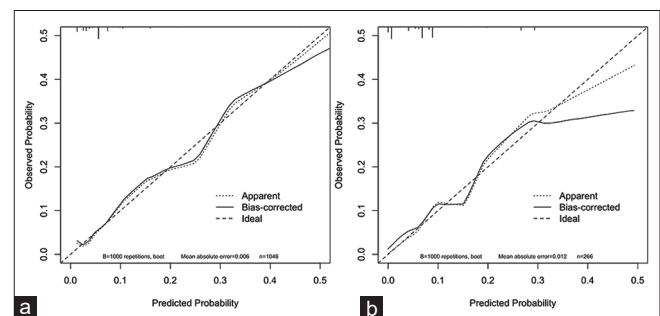


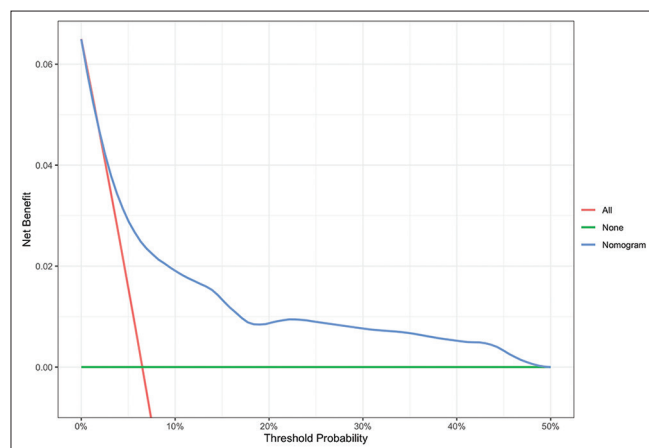
Figure 3: Calibration curve for predicting colorectal ESD-induced perforation. a) Derivation set, b) validation set

Table 2: Univariate and multivariate logistic regression analysis of risk factors for perforation in the derivation set

Variable	Category	No.	Univariate			Multivariate		
			Non-perforation group (n=978)	Perforation group (n=68)	P	Odds ratio	95% CI	P
Age in years	<65	523	489 (93.5)	34 (6.5)	0.830	1	-	-
	≥65	488	457 (93.6)	31 (6.4)				
Sex	Female	468	436 (93.2)	32 (6.8)	0.345	1	-	-
	Male	543	510 (93.9)	33 (6.1)				
BMI	<24	485	449 (92.6)	36 (7.4)	0.072	1	-	-
	≥24	526	497 (94.5)	29 (5.5)				
Tumor location	Rectum	286	272 (95.1)	14 (4.9)	0.190	2.96	1.46-6.19	0.013*
	Left colon	234	216 (92.3)	18 (7.7)		2.19	1.17-4.29	0.047*
	Right colon	526	490 (93.2)	36 (6.8)		1	-	-
Tumor size	<40 mm	838	801 (95.6)	37 (4.4)	<0.001*	1.66	0.92-2.99	0.155
	≥40 mm	163	132 (81.0)	31 (19.0)		1	-	-
Circumference of the lesion	<2/3	1035	972 (93.9)	63 (6.1)	<0.001*	7.18	2.18-23.44	0.006*
	≥2/3	11	6 (54.5)	5 (45.5)		1	-	-
Location with respect to colonic fold	Between folds	1019	962 (94.4)	57 (5.6)	<0.001*	3.69	1.63-8.18	0.007*
	Across folds	27	16 (59.3)	11 (40.7)		1	-	-
Preoperative biopsy	No	535	508 (95.0)	27 (5.0)	0.053	1	-	-
	Yes	511	470 (92.0)	41 (8.0)				
Location with scar	No	1015	951 (93.7)	64 (6.3)	0.152	1	-	-
	Yes	31	27 (87.1)	4 (12.9)				
Macroscopic type	Non-LST	379	368 (97.1)	11 (2.9)	<0.001*	2.23	1.16-4.49	0.050
	LST-G	343	310 (90.4)	33 (9.6)		2.34	1.22-4.67	0.036*
	LST-NG	324	300 (92.6)	24 (7.4)		1	-	-
Non-lifting sign	Negative	971	910 (93.7)	61 (6.3)	0.305	1	-	-
	Positive	75	68 (90.7)	7 (9.3)				
Submucosal fibrosis	F0/F1	875	835 (95.4)	40 (4.6)	<0.001*	2.52	1.51-4.16	0.003*
	F2	171	143 (83.6)	28 (16.4)		1	-	-
Tumor depth	Mucosa	947	883 (93.2)	64 (6.8)	0.302	1	-	-
	Submucosa	99	95 (96.0)	5 (4.0)				
Histological type	Adenoma	590	549 (93.1)	41 (6.9)	0.490	1	-	-
	Sessile serrated lesions	149	141 (94.6)	8 (5.4)				
	Carcinoma	307	288 (93.8)	19 (6.2)				
Endoscopist experience	<50 cases	383	361 (94.3)	22 (5.7)	0.490	1	-	-
	≥50 cases	663	617 (93.1)	46 (6.9)				
ESD type	Conventional ESD	280	258 (92.1)	22 (7.9)	0.284	1	-	-
	Hybrid ESD	766	720 (94.0)	46 (6.0)				
Procedure time	<60 min	795	770 (96.9)	25 (3.1)	<0.001*	1	-	-
	≥60 min	251	208 (82.9)	43 (17.1)				

SD, standard deviation; BMI, body mass index; LST, laterally spreading tumor; ESD, endoscopic submucosal dissection; * $P < 0.05$

1.51-4.16, $P = 0.003$), and LST-NG (OR = 2.34, 95% CI: 1.22-4.67, $P = 0.036$) were independent risk factors. The pre-operative observable factors with significant differences identified in the univariate analysis were

**Figure 4: DCA of the nomogram**

included in the multivariate regression analysis to obtain predictive factors for perforation. The results revealed the following predictors: Tumor located in the left colon (OR = 2.39, 95% CI: 1.20-4.87, $P = 0.040$), size ≥ 40 mm (OR = 3.36, 95% CI: 1.96-5.71, $P < 0.001$), $\geq 2/3$ circumference (OR = 7.55, 95% CI: 2.36-23.99, $P = 0.004$), located across folds (OR = 6.26, 95% CI: 2.82-13.64, $P < 0.001$), and LST (non-granular type, OR = 2.34, 95% CI: 1.25-4.64, $P = 0.029$; granular type, OR = 2.46, 95% CI: 1.32-4.81, $P = 0.021$) [Table 3].

Development and validation of the prediction nomogram

The predictive model incorporating these independent predictors was developed and presented as the nomogram [Figure 1]. Based on ROC analysis, the C-statistics in the derivation and validation sets were 0.750 (95% CI: 0.684–0.816) and 0.806 (0.717–0.894), respectively, indicating good discrimination of the

Table 3: Multivariate logistic regression analysis of independent preoperative predictors for perforation in the derivation set

Variable	Category	β	Odds ratio (95% CI)	P
Tumor location	Rectum		1	
	Left colon	0.869	2.39 (1.20-4.87)	0.040*
	Right colon	0.526	1.69 (0.92-3.26)	0.170
Tumor size	<40 mm		1	
	≥ 40 mm	1.213	3.36 (1.96-5.71)	<0.001*
Circumference of the lesion	<2/3		1	
	$\geq 2/3$	2.021	7.55 (2.36-23.99)	0.004*
Location with respect to colonic fold	Between folds		1	
	Across folds	1.835	6.26 (2.82-13.64)	<0.001*
Macroscopic type	Non-LST		1	
	LST-G	0.901	2.46 (1.32-4.81)	0.021*
	LST-NG	0.860	2.34 (1.25-4.64)	0.029*

* $P < 0.05$

nomogram [Figure 2]. The Hosmer–Lemeshow test presented a nonsignificant statistic ($P = 0.931$), which suggested that the predicted probability was close to the actual probability. Using bootstrap resampling, the bias-corrected lines closely aligned with the ideal lines, indicating a good agreement between prediction and observation in both the derivation cohort and the validation cohort for the calibration curve of the nomogram, for the probability of perforation [Figure 3]. The DCA showed that the predictive nomogram could be used to predict perforation and provide clinical net benefit if the risk threshold was between 0 and 50% [Figure 4].

DISCUSSION

The findings of the current study revealed that tumors exceeding 2/3 circumference, submucosal fibrosis, tumor location in the colon, LST-NG, and located across folds were identified as independent risk factors for perforation during ESD for the treatment of colorectal neoplasms. Tumor size ≥ 40 mm, located in the left colon, $\geq 2/3$ circumference, located across folds, and LST type were predictors for colorectal ESD-induced perforation. Notably, a major strength of our study lies in the development of the first user-friendly nomogram model for pre-operative prediction of colorectal lesion perforation, which exhibited satisfactory predictive accuracy and was based on a comprehensive and sizable database.

Although safety strategies and advanced equipment have improved, the significant challenge of high perforation risk remains a hindrance to the widespread adoption of colorectal ESD. The incidence of perforation during colorectal ESD ranges from 1.4% to 20.4% for lesions.^[10–13] This study further confirms the relatively high overall incidence of perforation, with rates of 6.7% in the derivation cohort and 8.3% in the validation cohort. In relation to the risk factors for perforation, large tumor size,^[10,15] colonic location,^[15,16,30] LST type,^[31] and submucosal fibrosis^[10,15] have been previously

described. This Current study further confirmed these findings. Despite our initial univariate analysis not revealing any significant influence of tumor location on perforation, based on clinical experience and previous literature, the rectum is considered the optimal site for ESD in terms of safety and ease with its thicker wall, stable position, absence of folds or bends, easy accessibility, and larger luminal diameter. Therefore, we hypothesize that the rectal location may possess protective factors against perforation and have included it in further multivariate analysis. The final results indeed demonstrate that the colonic location is indeed a risk factor for perforation (left colon, OR = 2.96, 95% CI: 1.46-6.19, $P = 0.013$; right colon, OR = 2.19, 95% CI: 1.17-4.29, $P = 0.047$). Tanaka *et al.*^[32] have previously reported that an increase in endoscopist experience is associated with a reduction in the rate of perforation during ESD. Hong *et al.*^[15] also reported that endoscopist's experience (<50 ESDs) is a risk factor for perforation. Nevertheless, our study did not find a correlation between endoscopist's experience and perforation, which could be attributed to the fact that difficult lesions in the real world are often entrusted to experienced endoscopists, and experts selected the appropriate lesion for non-experts. Another factor that has been largely neglected in previous studies is the procedure time. The time taken to perform the procedure is crucial in assessing the occurrence of complications, as a higher rate of complications may be associated with operator fatigue resulting from prolonged procedure time. On the other hand, challenges related to poor endoscopic operability and the precise localization of lesions in specific anatomical sites can contribute to prolonged procedure times, subsequently elevating the risk of perforation. The identification of prolonged procedure time as a risk factor in our study likely arises from the inherent complexities of ESD, including those influenced by colon flexure, submucosal fibrosis, and the intricacies of endoscopic maneuverability.

In our study, multivariate analysis consistently showed that a procedure time of 60 minutes or longer was an independent

risk factor for perforation. Prior research suggests a tumor with a circumference $\geq 2/3$ was the strongest risk indicator for colorectal ESD difficulty.^[33] However, no previous studies have investigated its association with perforation. Our study demonstrates that tumor $\geq 2/3$ circumference serves as a predictive factor for perforation in colorectal ESD. Managing such lesions during the ESD requires constant adjustment of the resection direction to achieve en bloc resection, which is both time-consuming and demanding, substantially heightening the risk of perforation. Limited research has been conducted on the relationship between the tumor's location with respect to the colonic folds and complications in colorectal ESD. A previous study revealed that polyps extending across the folds pose a risk for incomplete resection during EMR.^[24] Our study identified that tumors located across folds act as predictive factors for perforation in colorectal ESD.

Two prior studies have devised predictive models for colorectal ESD-induced perforation. Initially, Hong *et al.*^[15] developed a risk score model using coefficients from a multifactorial logistic regression model. This model incorporated four predictors: tumor size (1-cm increments, OR 1.20), endoscopist experience (≥ 50 ESDs, OR 0.59), colonic location (OR 2.20), and submucosal fibrosis (OR 2.00). However, submucosal fibrosis is difficult to predict preoperatively, and this model lacked prospective validation. Moreover, its risk stratification results (perforation rate: 4.1% for low risk and 11.6% for high risk) were deemed unsatisfactory, despite achieving AUC of 0.678 and 0.675 in the derivation and validation sets, respectively. Likewise, Imai *et al.*^[16] developed a scoring model based on logistic regression analysis that assigned risk stratification scores as low, moderate, and high risk, yielding corresponding perforation rates of 0%, 2.3%, and 25.8%. The model was successfully validated in a prospective cohort and exhibited AUC values of 0.73 and 0.80 in the derivation and validation sets, respectively. However, this model included predictors such as endoscopist experience, tumor location, morphology, scope operability, underlying fold, and fold convergence. The assessment of underlying fold and fold convergence is more challenging for less experienced endoscopists. To our knowledge, our study is the first study to develop a nomogram model for predicting colorectal ESD-induced perforation based on a large-scale derivation cohort. All the predictors included in the model can be readily obtained, even by inexperienced endoscopists. The model demonstrates good discrimination, with an AUC value of 0.750 in the derivation set. In order to account for the impact of technological advancements overtime on the predictive accuracy of our model, we further validated the model in a prospective cohort, achieving an AUC value of 0.806 in the validation cohort. We evaluated the

influence of our model on clinical decision-making through a comprehensive analysis of decision curves. DCA was utilized to evaluate the impact of nomogram-guided decisions on patient outcomes, providing valuable information on clinical consequences using a threshold probability approach. Net benefit, which considers the trade-off between true and false positives weighted by the harm associated with these results, was derived from DCA to assess the utility of the nomogram.^[34] In this study, the results demonstrated that when the threshold probability was set within the range of 0 to 50%, utilizing the nomogram for predicting the risk of perforation during colorectal ESD yielded greater benefit. By utilizing the predictors included in the nomogram, it is possible to calculate the individual risk of colorectal ESD-induced perforation. The user-friendly nomogram enhances the understanding of risk and facilitates enhanced clinical decision-making.

Several limitations of this study need to be addressed. First, although we conducted a prospective validation to establish the predictability of our model, further validation on a large-scale dataset would be necessary to strengthen its reliability. Second, our analysis was limited to a single tertiary center setting, and it is imperative to include multiple centers to broaden the population spectrum for external validation. Additionally, due to the retrospective design, certain factors such as paradoxical movement and endoscope operability, which have been identified as predictors for perforation,^[15] could not be collected and analyzed in the current retrospective design. However, our model, consisting of five readily available factors, demonstrates promising application potential.

In conclusion, tumor size ≥ 40 mm, located in the left colon, $\geq 2/3$ circumference, located across folds and LST type were independent predictors for colorectal ESD-induced perforation. The risk-predicting model incorporating these factors can quantitatively predict the risk of colorectal ESD-induced perforation preoperatively. Predicting the risk of perforation allows for comparing the risks and benefits of ESD, implementing preventive strategies, selecting the favorable therapeutic approach, and optimizing resource allocation.

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Conflicts of interest

There are no conflicts of interest.

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